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A Journal Devoted to Scientific and Practical Beekeeping.

ENTERED AT THE POST-OFFICE, SALER, AS SECOND-CLASS MATTER.

Published Monthly.

S. M. LORR, Publisher & Prop' r.

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WINTHAM, MASS., JUNE 15, 1885.

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AMERICAN APICULTURIST. VOL. III. NO. 6, JUNE 15, 1885 ***

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*BEEKEEPING AS A PURSUIT.*¹

BY ARTHUR TODD.

This subject may be regarded from two standpoints—that of the man who, with income assured from other sources, pursues beekeeping for its pleasure; and that of the man who, wishing to increase his slender income, or actually make an income, turns to beekeeping with a view to profit on the capital and labor to be invested. But, as to the latter are denied none of the pleasures enjoyed by the former, it is from the latter standpoint alone that I shall review the subject.

Beekeeping is, strictly speaking, a branch of agriculture, and many a farmer is to-day getting a greater return from his investment in bees than that received from any of his other stock; but right here I say that beekeeping as a pursuit has to-day become a “specialty.” The man who enters upon this pursuit (leaving the question of capital aside) must be one endowed with physical and mental ability; a man with open eyes and ears, one ready for emergencies, prompt to do what is necessary at once, and one who is not easily discouraged.

The physical ability is required because beekeeping demands real hard work—yes, back-aching work—not suitable to the sick ladies and gentlemen so often ill-advised to go into beekeeping. The mental ability is required to keep the beekeeper abreast of the times and its rapidly changing conditions. Beekeeping is now a science, a study, and the conditions which govern one season, or colony of bees, will be completely changed for the next. Every stage in the life of a colony of bees requires to be understood. There must be no “guessing,” and this will bring us to the cultivation of the habit of observation, and a disposition to hear all that one can upon the special subject.

Emergencies will occur needing heroic treatment, but the beekeeper with mind and hand trained by experience and thoughtful consideration of his “specialty,” will rise superior to any occasion, and when discouragement

comes, as it inevitably will, in the words of the immortal Longfellow, “He will look not mournfully into the past, it comes not back again, but wisely improve the future for it is his.”

Pleasure and profit go hand in hand, as a rule, in this specialty, although the former is not unalloyed by a liberal application of the “business end” of the little busy bee, and the latter by a recurrence of poor honey seasons. In nature are found both the beautiful and the sublime; in the hive both are constantly under the beekeeper’s eye, teaching him to look with amazement from “nature up to nature’s God.” As he views his hive and sees the city grow, and population increase, the waxen walls, and stores well filled, the free-born citizen hurrying to and fro, each with his special task, outside of the thoughts of profit will come to the most unimpressible, thoughts of wonder and admiration for the works of that great Architect of the universe who said, “Let there be life and there was life.”

The profits of beekeeping are what? To many a one they hold out the hopes of “the glorious privilege of being independent;” and to obtain these profits the specialist, gifted with the requisite mental and physical qualities, must be “the right man in the right place.” He must have hives of the movable-frame order. Moses Quinby wrote thus, in 1858: “There is not the least doubt, in my mind, that whoever realizes the greatest profit from his bees will have to retain the movable combs in some form;” and who of us will gainsay this to-day? Out of the many styles of movable-comb hives now in existence, the beekeeper will select one best fitted for the business in which he means to engage, be it the production of comb or extracted honey, queen-rearing, bee-selling, or a combination of all.

The specialist who intends to rear bees for sale will do well to employ that hive which will take the size and style of frame most in use in the district in which he resides. Interchangeability of parts is a grand secret of success, and the beekeeper who can sell a colony of bees, or buy a colony well knowing that each and every frame is usable in his own or his neighbors’ hives, has made a step in the right direction. The main points in a good hive are, “Simplicity of construction, combining plenty of bee-space with perfect ease of manipulation.”

The race of bees will next engage the specialist's attention. Study and experience, and also the actual line of business engaged in, will best decide this point. The black, the Italian, the Syrian, the Cyprian, and the Carniolan, alike have their votaries. At present, for all purposes of sale and honey-gathering, the Ligurian or Italian-Alp bee is the principal one in demand; but the very best race of bees will afford but little profit unless the queens are carefully looked after. As fast as signs of senility appear, these should be removed and their places supplied by younger and more vigorous queens. The apiarist for profit should not only rear queens, but know how, when and where to replace them. He should also know the requisites of a good queen, and how to judge of her progeny.

Pasture to the beekeeper is everything; if that be poor, his returns will be poor; hence he should carefully examine his location. Districts vary greatly in their flora, and by a careful study of this question before locating, disappointment will be avoided. The beekeeper should be a walking calendar of the flora of his neighborhood for miles around, then, as the honey comes pouring in, he can tell its source and label it accordingly. This knowledge will enable him to build up colonies, and follow the old advice, "Keep your colonies strong;" so that when the honey does come, there are bees to gather it in.

The management of bees kept for profit will vary according to the object of the beekeeper, whether it be the production of honey or the rearing of bees or queens. In running for honey alone, we have the swarming and the non-swarming methods. The experiences of good bee-men are so diversified that one is reminded of the old saying, "when doctors differ, the patient dies." The bee-man must strike out his own line of action suitable to his own special circumstances. In running for extracted honey, swarming is, to a great extent, controlled, for "Poverty maketh humble;" but I insist that the good bee-man will know the condition of each hive, and act accordingly.

The specialist is a man who reads, and although he may not get or use a single one of the many traps, or patent articles now offered, he should know all about them; for at any moment, what he has read about these things may give him an idea, the successful carrying out of which may help him over a difficulty. The capacity of the beekeeper to attend to a certain number of

colonies, be it greater or less, will have a great influence on the profits of the pursuit. As a pursuit, beekeeping should not be entered into without careful thought and consideration as to the capital required, the location and the suitability of the employment to one's temperament. To-day, before embarking in the business, it is possible for the intending beekeeper to serve an actual and willing apprenticeship in the yards of well-known and successful bee-masters. I need dwell not upon the advantages of this plan for they are obvious.

To the enthusiast with but small experience, I would say, "Go slow!" Read the good bee-literature now so easy to be obtained, and never be above learning from others. Visit beekeepers wherever you can enjoy the privilege, attend bee-conventions, and gradually a store of knowledge will be gathered upon which you will draw with profit later on.

Profitable beekeeping as a pursuit is, to my mind, the outcome of the union of two great factors—"talent" and "tact;" for "talent is power, tact is skill; talent is wealth, tact is ready money; talent knows what to do, tact knows how to do it; talent makes the world wonder that it gets on no faster, tact excites astonishment that it gets on so fast; talent may obtain a living, but tact will make one. Talent convinces, tact converts; talent is an honor to the profession, tact has the knack of slipping into good places, and keeping them; it seems to know everything without learning anything: it has no left hand, no deaf ear, no blind side, with a full knowledge of the Pythagorean doctrine, 'that a man ought rather to be silent, or say something better than silence.'"

I submit these remarks to my fellow beekeepers, being painfully conscious of many shortcomings from the high standard of excellence that man should attend to who in these days goes into "beekeeping as a pursuit."

Germantown, Pa.

HOWES' REVERSIBLE FRAME-SUPPORT.

BY C. J. F. HOWES.

To whom does the invention belong? From articles lately appearing in *Gleanings in Bee Culture*, and editorial comments thereon, I think there is a misapprehension of what the above invention consists, or what it really is, and whose property it is. The above-mentioned articles and editorials are, I feel, doing me an injustice, and have a tendency, virtually, to rob me of all the benefits, to say nothing of the "honors," of the discovery, which I had considered to be *my property*.

As to what constitutes my invention, I will quote from my article in the *A. B. Journal*, page 57, in reply to Mr. Heddon's claim to the invention of the frame illustrated in *Gleanings*, page 104.

"At the annual meeting of the Southeastern Michigan Beekeepers' Association, held at Adrian, Mich., Jan. 23, 1884, I exhibited samples of a device for reversing brood-frames, which device, or plan, suspended the frame by strips of wood, or metal, which strips were pivoted to the centre of the end-bars, and extended up to the top of the frame, there forming projecting arms to rest on the rabbets, and allowing the frame to revolve on these pivots.

In describing the device before the convention, I distinctly claimed as my invention, the plan of suspending the frame between side-strips pivoted to the end-bars, as described."

Previous to the illustration and description of my device, all reversible-frames had fixtures at both *top* and *bottom*; see *Gleanings* for 1882, page 71, also 1883, page 65, Burgess' device; 1884, page 155, Baldrige's device; and 1884, page 332, Hetherington's device. These attachments were entirely different in principle from the "Howes' Supports." No one had ever suggested revolving the frames on "*centre pivots*," previous to the illustration, and description of my frame in *Gleanings* for 1884, page 156.

Soon after I began to manufacture and sell Howes' Reversible Frame and Supports,—as advertised in *Gleanings*, for 1884, page 285,—reversing devices began to appear from all quarters; both men and women joining in the scramble for the “honor,” if not for the profits of the invention; each one suspending the frame by “*centre pivots*,” as I had described them. Several used hoop-iron bent at a right angle to form projecting arm. (Prof. Cook at the Michigan convention claimed to have tried this plan, though a lady, I think, first described it in *Gleanings*).

Mr. Root has often, through *Gleanings*, acknowledged that these different devices are, practically the same thing as the “*Howes' Support*,” still he does not hesitate to manufacture, advertise and sell them, as if the invention was common property. (See editorial remarks in *Gleanings*; page 74, describing Mr. Nuzvinis' device; then Editorial in next number, page 104, on the same subject.)

I submit the question, in all seriousness. Does not this state of things *justify* anyone, in securing his rights to the labor of his brain, by a *patent*, as provided by law? If a better device, to secure the results aimed at, by reversing the brood-combs, shall be discovered, I shall be glad to adopt it, in my own apiary, and will, willingly, pay for the privilege. Until then, I request all to “please keep off my preserve.”

Adrian, Mich., Feb. 7, 1885.

WHICH IS THE MOST PROFITABLE RACE OF BEES.

BY D. F. LASHIER.

Which is the most profitable race of bees regarding brooding, comb-building, honey-gathering disposition and wintering qualities?

I have no desire to injure anyone's business and wish merely to bring to notice a few facts which it seems to me to have been overlooked.

Perhaps a few notes founded upon years of experience would not be out of place, especially as my motives are entirely unselfish.

I commenced beekeeping in 1872 with one colony of gray bees in a box hive. I purchased this hive of a neighbor whose bees seemed to be very hardy and gentle. They are of uniform size and as large as any Italians that I ever have seen, even when the latter were reared in comb of their own building. I hive all my gray bees without any protection whatever and the same when looking for queens, etc. With the Italians this would be perfect madness.[2](#)

Doubtless some will say "your gentle bees must suffer from being robbed." Not so, my friends, they are good protectors of their homes.

I have purchased queens from some of the best breeders in the States hoping to get the best in the market, and I never have, as yet, seen any that will hold their own with my gray bees.

In breeding, the Italians commence a trifle earlier in spring, but they dwindle so badly that when fruit trees bloom they are not as strong as the gray bees.

For honey-gathering from the white and red clover the Italians and gray bees are about equal, but when buckwheat is in bloom the grays beat the

Italians by fifty per cent.

I have wintered both races indoors and out of doors. The grays seem to become dormant not caring to move about, while the Italians are uneasy, crawling out of their hives and wasting away.

My gray bees have steadily increased by natural swarming³ from the one colony to 120, all in the same apiary; giving me, in an average season, a nice surplus of box honey, and in a very poor season holding their own without feeding or spring dwindling.

I think that had one-half the pains been taken to improve some of our native bees that have been devoted to rearing foreign races, beekeeping of to-day would be in a far better condition. It is the general result and the general summing up that decide which is the more profitable vocation.

Of late years I have wintered my bees in a frost-proof building, and have found it to be a great saving of honey. At some future time I will tell your readers, if they wish, how this building is constructed so as to carry bees safely through five months of as cold winter weather as ever existed in my section, together with my experience in fruit raising in connection with beekeeping and how I manage to save my natural swarms from absconding.

This having swarms decamp to parts unknown is all wrong. I have had more swarms come to me than I ever had desert, and the idea that bees injure fruit blossoms is altogether erroneous. Why! we were obliged to prop up our plum trees last season to prevent them from breaking down with the load of plums, and of cherries we had a most bountiful crop and this right in our apiary too.

I should be pleased to give your readers a paper on fruit and bees if it would be acceptable.⁴

Hooper, N. Y.

HOW TO FORM NUCLEI.

BY HENRY ALLEY.

Most beekeepers wish to rear a few queens and must have for such a purpose a few nucleus colonies.

The following plan for forming them is an easy and simple one. The nucleus hives should be constructed of light material and about one-third the width of the large hive, and the covers should be cleated to prevent splitting and warping. If the hives are painted they will last many years.

The nucleus colonies should consist of three combs and two quarts of bees. When ready to form them, place in one of the hives two combs containing honey and, in the centre, one containing brood, after which add the bees. Perhaps the better way would be to remove from a full colony a comb containing brood, together with the adhering bees, being careful of course not to take the queen with them. In such case there will be a sufficient number of bees on the comb to care for the brood; if not, a few more may be brushed from another comb into the hive.

After the combs and bees have been placed in the hive, confine the latter to the hives for thirty-six hours and release them early on the morning of the third day.

Do not remove the screen (with which they have been confined) from the entrance except just before dark or early in the morning; as, if released in the middle of the day, the bees would rush out and many would not return.

After having been confined in the hive for thirty-six hours, the bees will have constructed several queen cells and when released will return to the new location.

A matured queen cell may be given them at this time; or, when they have been queenless seventy-two hours, a virgin queen can be introduced safely.

Before releasing the bees the nuclei should be placed some distance from the stands from which the bees were taken. While the bees are confined in the hives they should be supplied with water. For such purpose I find the cone feeder very useful.

If the reader has studied the article in the May number in reference to the drone trap he is prepared to have his queens purely mated with any particular strain of drones in the apiary.

Wenham, Mass.

A GUIDE TO THE BEST METHODS OF BEEKEEPING.

BY J. L. CHRIST.

R. F. Holterman, Translator.
(Continued from p. 90, Vol. III.)

PERTAINING TO THE FINEST AND BEST COLONIES.

It is a principle in beekeeping if one desires to derive a benefit from his bees, to see that one keeps very populous colonies. The mere number of colonies has nothing to do with amount of value; but their strength, the number of inhabitants in a hive, is the measure of its worth. One single populous colony is worth more and will store more honey than four weak ones. In fourteen days the one will bring in more honey than the four will in four weeks.

I place a strong colony at 40,000 working bees; of these 13,000 to 16,000 can daily fly out and bring in stores; the remainder stay at home to care for the brood, to build comb, and to perform such other duties as may be required of them.[5](#)

Of four weak colonies, however, each calculated at 12,000 workers, only 4,000 can fly out leaving 8,000 at home. These four colonies together not only cannot send to the fields as many workers as the one strong one, but they also labor under many disadvantages.

It may be good weather for eight days and the flow of honey abundant and the strong colony may in that time gather all its winter stores, but the weak ones can take only sufficient advantage to gather at most one-fourth of the required stores. If unfavorable weather should follow, and the flow of honey cease, the strong colony is supplied and the four weak ones are lost in the winter if they be not fed, which latter is associated with much expense, trouble and inconvenience and even then often fails, not to mention the

facts that the weaker ones cannot depend upon themselves as well against robbers, moths, ants, etc.; and in winter they cannot maintain the proper warmth as well, are more liable to be frozen, and are less able to stand the changes in temperature.

They cannot rear brood as early as the strong one and there are many advantages the strong one has over the weaker, one of the most prominent of which is that the strong one displays more energy and is more industrious than the weaker.

IN THE ORDINARY STRAW SKEPS ONE CANNOT CARRY OUT ONE'S DESIRES COMPLETELY IN MANAGING BEES.

As important and well known as the fact now is of having the colonies strong, one cannot make them so if they are kept in the common, simple straw basket where one does nothing but destroy in a slovenly way, especially farmers. In the fall they take the heaviest and lightest colonies and in a sinful and thoughtless manner kill and smother its inhabitants thus doing themselves a deliberate injury, as if they permitted these useful creatures, these patterns of industry, to live, they would gain far more.

I once saw a beekeeper take a very heavy colony consisting of two colonies which in swarming clustered together, and smother them, because he thought that owing to the large number of bees the colony might not have enough winter stores. Yes! a clown of a fellow actually burned with straw his young swarms, because they came rather late. But I do not intend to occupy my time describing the wrong mode of keeping bees, as through the length and breadth of the land this has been so passionately spoken of and they will learn, only as matters progress, to adopt a better mode of beekeeping.

THE MAGAZINE STOCK⁶ ARE CONVENIENT, PARTICULARLY WITH THE WOODEN TOP BOXES WITH A GLASS SIDE.

If one wishes to build up populous colonies, one must commence by controlling swarming: namely, swarming often. To do this, one must

provide roomy dwellings and those that can be enlarged gradually; without this the object would fail. If one should give the bees a large dwelling at once they would become discouraged and would not half fill the hive and there would be many other disadvantages.

The dwelling must also be arranged in such a manner that the bees can be handled with ease and without damage to the bees, or ever to have to destroy the latter to enjoy the product of their industry and control their surplus in honey and wax.

All this is required; but now as to the care of the “magazine.”

As the bees conduct their domestic affairs within a limited space and they from time to time according to the demands of time, attentions, etc., are increased or diminished, one generally makes them of straw (at least I have seen no others) and lathes which are very useful, if not too large (as they generally are); nevertheless, these straw “storing cases” have several drawbacks which I have found by observation and manipulation. Thus, some years ago, I conceived the idea of making, as far as possible, those that were more complete and convenient; to that end I made wooden four-cornered hives of boards and put in the same at least one pane of glass which, although only costing but little, is of inestimable value.

I improved on these until I found the most useful and convenient to handle.

The samples that I have I not only had myself for several years, but I also made some for good friends, and others made copies of them and their great value makes me recommend them unhesitatingly. These hives are very little more expensive than straw (if they are made plain and cheap), they last longer, are better and more convenient and can be made anywhere, while men who can make straw hives are often difficult to get.

One should not allow himself to be frightened into thinking they are too expensive, when conducting an extensive bee business; or that, if one begins with them, the profits will soon disappear.

I will just describe their completeness and their general utility and their advantages over the straw hive, more especially for the purpose of giving

guidance how to make them of the greatest use in beekeeping.

Rodheim, Germany, July, 1783.

[To be continued.]

IS BEEKEEPING PROFITABLE.

BY T. F. ARUNDELL.

My attention being called to the article headed, "Is Beekeeping Profitable" (page 64, March No.), I will answer it to the best of my ability.

The answers to questions 1 and 2 depend simply on the annual yield of honey per colony, and the success attained in wintering. Should a surplus of 100 lbs. or more be obtained from each colony (spring count), together with a moderate increase in bees and with little or no loss in wintering, success would be assured; but, should the average yearly surplus amount to no more than 40 or 50 lbs. coupled with severe losses in wintering, the business could not prove otherwise than a financial failure.

No. 3. In this section one man can properly manipulate at least 150 colonies and their increase, make the extra hives, and extract all the honey from them, excepting possibly, a very few days when honey is stored more rapidly than usual.

As to tools wearing out with use, I will say that I have extracted 40,000 lbs. (20 tons), in the last two summers with a four frame "Novice" geared extractor with no appreciable wear, and my honey knives are as good as when first bought. A well made hive should last a lifetime, so that the depreciation in value of utensils is more imaginary than real.

No. 4. Given a *properly arranged hive* and appropriate tools a man can extract from 120 to 140 lbs. per hour, say 1200 lbs. per day.

In four successive days last July I extracted and filled into cans 5,200 lbs. of honey, working about eight hours each day extracting and filling cans from the tanks mornings and evenings. Hives in my apiary average about 40 lbs. each extracting.

No. 5. In regard to the number of colonies an apiarist could oversee, it depends more on the talent and genius of the overseer than upon the number of colonies possessed. While one man can direct and control hundreds of employés, another cannot even manage his own labor to lead to the best results.

At some future time I would like to describe, in your Journal, the best and speediest methods of extracting honey in large apiaries as practised in this country.

Santa Paula, Cal.

INSTRUCTIONS TO BEGINNERS.

BY THE EDITOR.

There are, properly speaking, two systems of keeping bees: the one adapted to the needs of the expert and specialist who keeps a large number of colonies and devotes his entire time and attention to the pursuit; the other adapted to an apiarist who keeps bees in connection with some other business, either for the pleasure and recreation, or for the purpose of adding another source of income.

The latter class constitute the majority of the beekeeping fraternity and only too often are their needs overlooked by those who write upon the subject of apiculture.

While we are aware that if one is adapted to beekeeping, and enters into it properly in a favorable location and masters it that success will follow; yet, as a rule, we advise keeping bees in connection with some other vocation, as when one becomes thoroughly familiar with all the requirements of beekeeping it is an easy matter to enlarge his apiary. Then, again, a few colonies properly managed will generally give far better results than can be obtained from a large apiary.

Those who are just commencing should remember that the less they handle their bees, and yet accomplish what may be required, the better for their colonies.

If one is naturally nervous, it is best to wear a bee veil at first and when manipulating the colonies work gently and avoid jarring or fretting them.

When looking for the queen, blow a little smoke in at the entrance as this causes the queen to run up on the comb and the bees that may be running about on the bottom board will fill with honey.

One of the first steps for the beginner is to decide, as far as possible, to which class he intends to belong, and what amount of capital he can safely invest.

Where one has a limited capital and wishes to become an expert or a specialist, it is far better to begin on a small scale and gradually increase his number of colonies making them pay their way and also furnish funds for new investments.

If your first lessons have been gleaned from flaming advertisements or reports of enormous yields, or through reading some of the overdrawn works on apiculture (so written with the purpose of making new converts), just take some wholesome, practical food for study and thought, both by securing one or more of the works on practical apiculture mentioned in this journal, and by visiting some practical and successful apiarist. In this way, you will be prepared to look at both sides; and if, after doing this, you enter into beekeeping with a determination to succeed you are certain to make it pay, provided you are adapted to the business, and other things are equal.

There is not the slightest reason why nearly every person who has a fair-sized garden should not keep a few colonies of bees and thus provide the table with nature's purest and most healthful sweets. Success in any vocation always means hard work, together with push, tact, and energy. Thousands embark each year on the sea of business enterprise and the shoals and quicksands are strewn with stranded wrecks, yet there are those who, by rigid economy and shrewd management, accumulate a competency besides establishing a good remunerative business.

Our advice to those who wish to engage in beekeeping would as a rule be this. If at present you have no location, look about you and find a small place of from one to ten acres according to your means and the situation.

It is better to have the land slope to the south and east if possible and it should be well protected from the cold north and west winds. Perhaps you can rent or lease a place adapted to your needs. The surrounding country should be well supplied with bee pasturage in the shape of orchards, clover, basswood (if possible), wild flowers or many others that we might name but which are described in most of the works on apiculture.

Where one is located in the city or village and means to keep only a few colonies this advice is unnecessary, but with all others it is imperative that they locate in a good honey-producing district.

It is also best to learn if there are many bees kept where you wish to locate; as, while there is no law to prevent your establishing an apiary by the side of your neighbor, yet the latter has rights which it is proper and just to respect. This again will not matter without you intend to build up a large apiary.

One may secure a large yield of honey and yet find a poor market for it; hence it is always best to take into consideration the advantages for establishing a good home market. It will pay far better than shipping to large markets and giving all your profits to commission men.

There are so many items of interest which should serve as an introduction to these papers that we hardly know where to stop and must be necessarily brief and even leave many of them until we write again. In purchasing bees it is best if you want but from one to five colonies to purchase them of some reliable dealer and always select a standard frame, and it will pay you well to look into the merits of the various frames before making your purchase.

While for some reasons we prefer a frame about 10×15 , yet as the "Langstroth Standard" is now so largely in use and is no objection as regards wintering the bees, we deem it best to adopt it in our own apiaries.

Circumstances must in a great measure control these matters, but whatever style is adopted it should be adhered to, else much trouble and expense will result.

We deem the tenement hive the best for all purposes. True, the first cost is somewhat greater, but in the end it pays.

The hives should be constructed in as simple a manner as possible, and while if one wants but a few he can make them after obtaining his colony of bees and estimating the size of the brood chamber; yet it is much better if he wants five or more hives to purchase them in the flat.

While we prefer for working bees a cross between the Italian and Holylands (from Syria), yet as a rule we would recommend the Italian as the *best* for the average beekeeper. Experience will teach one which is the best race.

We shall endeavor to give illustrations of different styles of hives in our next paper but have been too busy to attend to it this month.

Our first advice is “Make haste slowly,” but “stick to it” until you have either mastered the business or found that you were better adapted to some other vocation.

NEW OBSERVATIONS ON THE NATURAL HISTORY OF BEES.

BY FRANCIS HUBER.

(Continued from p. 95, Vol III.)

Solicitous to learn its origin, and conjecturing that it might be masculine matter, he began to watch the motions of every drone in the hive, on purpose to seize the moment when it should be received by the eggs.

He assures us, that he saw several drones insinuate the posterior part of the body into the cells for that purpose. After frequent repetition of the first he entered on a long series of other experiments.

There was something very specious in this explanation: the experiments on which it was founded seemed correct; and it afforded a satisfactory reason for the prodigious number of males in a hive. At the same time the author had neglected to obviate one strong objection: larvæ appear when there are no drones.

From the month of September until April, hives are generally destitute of males; yet, notwithstanding their absence, the queen then lays fertile eggs.

Thus the prolific matter cannot be required for their impregnation, unless we shall suppose that it is necessary at a certain time of the year, while at every other season it is useless.

To discover the truth amidst these facts, apparently so contradictory, I determined to repeat Mr. Debrow's experiments, and to observe more precaution than he himself had done. First, I sought for that matter which he supposes *the prolific* in cells containing eggs. Several were actually found with such an appearance, and during the first day of observation, neither my assistant nor myself doubted the reality of the discovery. But we afterwards found it an illusion arising from the reflection of the light for nothing like a

fluid was visible except when the solar rays reached the bottom of the cells. This part is commonly covered by shining fragments of the cocoons of worms successively hatched, and the reflection of the light from these when much illuminated, produces an illusory effect. We proved it by the strictest examination for no vestiges of a fluid were perceptible when the cells were detached and cut asunder.

Though the first observation inspired us with some distrust of Mr. Debraw's discovery, we repeated his other experiments with the utmost care.

On the 6th of August, 1787, we immersed a hive, and with scrupulous attention examined all the bees while in the bath. We ascertained that there was no male, either large or small, and having examined every comb we found neither male nymph nor worm. When the bees were dry we replaced the whole, along with the queen in their habitation, and transported them into my cabinet. They were allowed full liberty; therefore they flew about and made their usual collections; but it being necessary that no male should enter the hive during the experiment, a glass tube was adapted to the entrance, of such dimensions that two bees only could pass at once; and we watched the tube attentively during the four or five days that the experiment continued. We should have instantly observed, and removed any male appearing, that the result of the experiment might be undisturbed, and I can positively affirm that not one was seen.

However, from the first day which was the 6th of August, the queen deposited fourteen eggs in the workers' cells; and all these were hatched on the tenth of the same month.

[*To be continued.*]

EDITORIAL.

Doubtless our friends will not take it amiss if we give but a brief editorial this month. Our new enterprise is receiving such hearty endorsement from every quarter that we must conclude that we have taken the right step after all. We only ask that the reader and our customers bear patiently with us for a while. The season has been so backward that we are in a rush but shall be amply able to meet all demands.

We have not utilized the question and answer department this month as the amount of work that we have been obliged to perform has been simply enormous, but we shall soon have that department in running order again.

We would urge upon our readers the importance of creating a home demand for honey this season in order that we may not have a glutted honey market with which to contend.

Work hard at the State conventions to have county associations formed; these are the educators that will prove the gateway to a home demand.

Prepare to make first-class exhibitions at county fairs next fall: it will pay. It may seem as though this advice was premature, but we have none too much time to prepare for these fairs.

Remember that if we ever have a permanent market for our honey the beekeepers must make it.

We wish our readers to come to us with their experiences, successes and failures and we shall endeavor to give them such instruction as will aid them in their work.

Do not fail to send for a few of our "Companions" and distribute them for us; it will be but little trouble and will aid us wonderfully in building up our subscription list.

We are preparing to give our readers a large number of illustrations and as our list increases we shall be able to add many new and interesting features to our Journal.

We have already sent out 5,000 “Companions” within the last two weeks, and the call is so great that we are printing 5,000 more.

EXPERIMENTAL BEE FARM NOTES.

On account of the many duties devolving upon us in establishing our bee farm we have not as yet been able to carry out any of the experiments that we mean to test for the benefit of our readers.

While for the purpose of experimenting we shall devote some colonies to the production of honey, both comb and extracted, yet this season's operations will be confined almost exclusively to queen-rearing.

We start with about one hundred colonies of the different races of bees. The season thus far (in this locality) has been unusually backward and trying, giving us but little warm, pleasant weather and but few days when the bees could gather honey freely.

Last winter most of our colonies were wintered in the beehouse and were removed to the summer stands, about the 28th of March, in fine condition and there seemed to be but little difference between their condition and that of those wintered on the summer stands. The first work of the season commenced when the bees were removed from the beehouse. The hives were cleared (as much as possible without removing the combs) of dead bees; the honey boards with which the hives were covered during winter were removed and replaced with mats on the top of which chaff cushions were placed.

We kept our colonies well packed that they might be snug and warm even during the sudden changes in temperature and we make it a rule never to remove such packing until warm weather has "come to stay."

As soon as the bees could be handled safely the colonies were examined and all combs outside the cluster were removed. This is an excellent practice, as by reducing the capacity of the brood chamber to the size of the cluster we enable the bees to maintain sufficient heat to carry on brood-rearing much more rapidly than when they are surrounded by a cold vacant space.

At the time of setting out the bees, the entrances to the hives were all contracted to about one inch and it is wonderful to see what a help this is to the bees in keeping the hive warm.

In a few days after being removed from the bee-house the bees were busily engaged in carrying in the artificial pollen (wheat flour) which we provided for them placing it near by in the apiary.

Brood-rearing soon commenced and in about four weeks the colonies began to show marked signs of increase.

Soon we found it necessary to add empty combs wherein the queens might deposit eggs. While in the hands of the expert this is a safe and sure method of building up the colonies rapidly, yet it must be conducted with caution else during a warm spell the queens may utilize more combs than the bees can cluster and thus neglect the brood in the outside combs which in such case must perish. Never spread brood faster than it can be covered and well protected by the bees even during the cool nights. During the last few days in April the weather was favorable for honey gathering and a few pounds were stored by each colony from the soft maple bloom.

Our colonies have been so well protected that they were not troubled with spring dwindling and soon they became so strong in numbers as to be in fine condition for queen-rearing.

It is poor policy both for the breeder and for the honey producer to attempt to rear queens with any but the most populous colonies if they want first-class queens.

At this date we have between 300 and 400 queen cells in all stages of construction and quite a large number of queens ready for fertilization.

Our queen-rearing is conducted by the methods given in the *Beekeepers' Handy Book*; and, indeed, after having practised all the various methods, we would as soon think of going back to the old box-hive system of keeping bees as to practise the old unsatisfactory and uncertain methods of rearing queen bees.

It is a pleasure to examine the nice evenly built and conveniently spaced rows of cells found in the queen-rearing colonies now under Mr. Alley's supervision. It is indeed a most interesting sight and one that we would be pleased to share with our beekeeping friends to whom we extend a most cordial invitation to visit us. We shall endeavor to impart to all our visitors all the information possible regarding queen-rearing and the general management of the apiary as conducted at our "bee farm." At present we have four races of bees from which we are propagating queens, prominent among which are the orange-yellow Italians. We confidently assert that we never saw or possessed a more beautiful or hardy strain of pure Italian bees.

We shall run four separate apiaries situated about three miles apart in order to keep each race strictly pure.

We have set out one hundred "prickly comfrey" plants, purchased of Mr. Arthur Todd of Philadelphia, Pa., and shall refer to them again later in the season. We have also sown one and a half acres of Bokhara clover for the bees and it will pay our readers to utilize every waste spot (at least) with either or both of the above. Increase in pasturage means increase in surplus honey. It is now time to prepare for the coming honey harvest and indeed in many sections of the country the surplus boxes have been placed in the hives, or the honey extractor resorted to. When the colonies that are to be run for section honey become populous and begin to build white comb along the edges of the top bars, and perhaps between the combs, it is well to place one set of sections on the hives but they should have only the amount of surplus room that they can utilize and other sections should be added as they are needed.

A piece of comb foundation one-half the size of the section, cut in the shape of a triangle and attached point downward, will be a wonderful help to the bees and prove a paying investment to the apiarist.

If some colonies seem strong but do not utilize the sections readily, just exchange their empty sections with partially filled sets (bees and all) from other colonies. This is a plan practised more than twenty-five years ago by Mr. Alley and Mr. John J. Gould, formerly of Wenham. Mr. G. was at one time one of the largest beekeepers in this State.

Mr. Pond, however, credits a beekeeper in Maine with being the originator of this most excellent plan for inducing the bees to enter the sections.

There is one disadvantage in connection with this practice that Mr. Pond and others fail to give.

It often happens that when a colony at work in the sections is disturbed, the queens will run up into the latter and in the removal may be lost. To prevent this make as little disturbance in the transferring as possible and smoke down, into the hive, all the bees that cluster on the tops of the frames.

The bees removed with the sections will not quarrel with their new neighbors.

There are many items that would prove interesting to our readers but we have already devoted more space than we intended to this department and must await another opportunity.

CORRESPONDENCE.

THE HONEY EXTRACTOR.

FRIEND LOCKE:

Your beautiful book, Vols. I and II of the “American Apiculturist,” is before me and its contents noted. It seems to fill the bill, being replete with items of importance especially to the more advanced apiarist. While we cannot forget the labors of a Langstroth, a Quinby, and a few other dear names, your work fills a more modern want, and if you keep in view the interest of honey producers, as I do not doubt you will, you ought to meet with success. I have been in the business of raising and marketing comb-honey about 35 years. Made a little at it once, but of late years it hardly pays. The injudicious use of the honey extractor has been a great damage to us as beekeepers, and it is the only thing I know of that can injure your success. This engine bee business is a mistake. Some are clamoring for a large iron extractor, that will extract not less than four combs in no time. I suppose their reason for it is that the larvæ will not expire under the operation. Some do not relish larval honey anyway, and Joseph prefers quality to quantity, nor does he believe it benefits brood or comb to be violently whirled in the vortex of death.

They talk about a glutted market. Well! well! honey is down and the mourners go about the streets. What shall we do? Create a home demand? I think friend L. C. Root has answered this question on page 48, Vol. II, “American Apiculturist” when he says “we must devise some means of producing smaller crops,” and I would add without diminishing our income that we annihilate the extractor. What other method can we rationally adopt to curtail the quantity and enhance the quality and thus keep up a demand at home and abroad? Self interest for one is self interest for all honey producers. It is better to make a little pay than a good deal not pay. I am not talking for the supply business; if I were, I should undoubtedly advocate the extractor, because what bees it does not kill in the larval state are hastened

to an untimely death by a cruel and barbarous system, which creates a demand for fresh victims, fresh queens and fresh workers, so that now the supply dealer alone makes anything.

When a law is passed by the Legislature against the wicked practice of extracting honey as now carried on, it will be the happiest day that the bee, or the beekeeper who keeps bees for profit, has seen. Of course I do not hope to do justice to this question in so short an article, but would submit it to your thoughtful consideration as one of the most important questions touching the interest of the American beekeeper.

Be gentle with the little bee
Which toils the summer day,
To bring its treasures to your home
From every hidden way.
The fragile insect needs the care
And kindness of your heart,
If you would win his services
To aid you in a start.

J. C. CLARK.

Alden, N. Y., Apr. 15, 1885.

CHAFF HIVES AND DESTRUCTION OF BROOD.

MR. EDITOR:

I drop these few lines to you to ascertain whether the “Apiculturist” is opposed to publishing opinions of beekeepers in regard to certain hives, in certain localities, their good and bad qualities, etc.

I sent A. I. Root my opinion of the chaff hive, in my locality, how I lost my bees very nearly all one season, by using them; and I proved the cause to be the fault of the hive.⁷ I must have been right, or he would certainly have published the article. At least I thought I was, since he didn’t publish it, which is about two years ago. Now he ought not to have been opposed to

publishing my article, because I have already read in the *Gleanings*, where he said he was always glad to have the children write pieces for it, because, as he says, they generally speak their mind right out, no matter whom it hits.

Now, I am no child, nor am I a professional writer; but I consider that my article was as well gotten up as any child could get one up, and it was just as plain spoken too, as any article a child ever wrote for *Gleanings*, but it must have hit in the wrong place.

Now, Mr. Editor, if the “American Apiculturist” is conducted and run on such selfish motives, I would rather have my money returned than be considered a subscriber.

You are at liberty to publish this if you choose.

Sometime since I saw an article in one of the bee journals, in answer to an inquiry as to the reason why a certain gentleman’s bees tore down their worker brood and dragged it out.

The answer that he received was this: in case the honey flow ceases suddenly, the bees will tear down their worker brood and rid themselves of it to prevent starvation which, so far as my experience goes, I have never found to be the case, although I have known the honey flow to cease very suddenly and very often with me.

When the honey flow suddenly ceases it is a very common thing to see the bees dragging out drone brood; but whenever I see any of my bees carrying out their worker brood I know that that colony is troubled with the moth (or wax) worm.

I then open the hive and assist the bees in ridding themselves of the pests, and I have never yet failed to find the latter travelling along just under the cappings of the brood, where their presence is easily detected by white streaks which are easily followed with the head of a large pin or the point of a knife and the worm removed without injuring the brood in the least.

I suppose that the bees, in their efforts to secure and remove the worms, tear out some of the worker brood.

J. C. SIMMONS.

Pottsville, Pa.

NEWS FROM NEW YORK.

FRIEND LOCKE:

Vols. I and II of the "Api" (bound in one) came to hand in nice shape. Its make-up is far superior to bound vols. of papers and magazines in general, being free from advertisements through the body of the book. In fact one could hardly tell that it was a magazine as it has a strong appearance of being compiled expressly for book form. The mechanical part of the work is ahead of any volume on apiculture we know of, and of course the literary part is *par excellence*, coming as it does from such men as L. C. Root, A. J. Cook, J. E. Pond, Arthur Todd, etc., etc.

No apiarist who wishes to be up with the times can afford to be without it. It is an ornament to any library.

The snow is all gone, but with the exception of a few days the weather has been cold and May 1 and 2 it froze quite hard in the night. The winter has been severe. The loss in bees in this county is fully 50 per cent, yet what are left are in fair condition.

Clover appears not to have been killed during the winter and I think we may look for a fair crop of honey if we can get our stocks strong enough to swarm in time.

Wishing you success, I remain yours truly,

C. M. GOODSPEED.

Thorn Hill, N. Y., May 4, 1885.

NEW YORK AND FLORIDA.

DEAR SIR:

Books, papers and "Apiculturist" received. Thanks.

Owing to the continued cold weather in January and February the orange bloom is not at its prime yet at this date (March 11), so I will omit the report of its value as a honey producer until next month.

The temperature for February was yet lower than for January. I have taken three observations daily for the two months.

For January, at 7 A. M., average for the month 55.° At 1 P. M., in sun and cloudy weather, highest 78°, at 8 P. M., 54.° Extreme temperature, hottest, 98,° coolest 34.°

For February, 7 A. M., 47,° 1 P. M. in sun, 79,° 8 P. M., 53.° Highest above 92,° lowest, 28.°

Greatest variation during seven hours, 58.°

February at 7 A. M., temperature only 7° colder; at 1 P. M. 1° hotter and at 8 P. M., 1° colder.

January was very foggy and damp and much more uniform in temperature than February.

Bees work every favorable day but no swarming in this vicinity this month.[8](#)

C. G. FERRIS.

Sanford, Fla., March 11, 1885.

EXCHANGES.

LIMITATION OF THE VISUAL FIELD OF THE WORKER HONEY-BEE'S OCELLI. BY THE REV. J. L. ZABRISKIE.⁹—The honey-bee is a remarkably hairy insect. On the head the hairs are dense, and of various lengths; and they cover every part, even the compound eyes and the mandibles. The antennæ, however, are apparently smooth, having only microscopic hairs; and a path through the long hairs, from each ocellus, or simple eye, directly outward,—to be described more fully presently,—is practically smooth.

The ocelli are so situated that when the bee is at rest and the face vertical, they are directly on the top of the head, arranged as an equilateral triangle, and one ocellus is directed to the front, one to the right side, and one to the left¹⁰.

Long, branching hairs on the crown of the head stand thick like a miniature forest, so that an ocellus is scarcely discernible except from a particular point of view; and then the observer remarks an opening through the hairs—a cleared pathway, as it were, in such a forest—and notes that the ocellus, looking like a glittering globe half immersed in the substance of the head, lies at the inner end of the path. The opening connected with the front ocellus expands forward from it like a funnel, with an angle of about fifteen degrees. The side ocelli have paths more narrow, but opening more vertically; so that the two together command a field which, though hedged in anteriorly and posteriorly, embraces, in a plane transverse, of course, to the axis of the insect's body, an arc of nearly one hundred and eighty degrees.

These paths through the hairs appear to me to be indications that the ocelli are intended for distant vision, although the opinion that near vision is their function is held by eminent opticians.

The ocelli are nearly hemispherical, and the diameter of each is about fifteen times that of a facet of the compound eye. Such a form of lens

would, I will concede, indicate for these organs a short focus, and hence, a fitness for near vision.

But if the ocelli are intended for near objects, it is difficult to understand why they are surrounded by a growth of hair so dense as to permit unobstructed vision only in a very narrow field, and why they are so placed on the top of the head as to be debarred from seeing any objects in the neighborhood of the mandibles and the proboscis, the ability to see which objects would appear to be very necessary in the constant and delicate labors of the worker honey-bee among the flowers.

Dr. Zabriskie exhibited the head of the worker honey-bee for the purpose of illustrating the above remarks. At the conclusion of his observations, he added: “Besides the worker honey-bee, I have brought for exhibition the drone and the queen of the same species, and the queen-cells; the queen of *Bombus Virginicus*, one of our native humble-bees; the *Melissodes binotata*, male and female; the *Melissodes pruinosa*, both sexes; the beautiful *Anthophora dispar*, male and female of Tunis, Africa; and the celebrated little stingless bee of Abyssinia, the *Trigona Beccarii*, which lives in immense colonies, and stores large quantities of honey. The *Anthophora dispar* has a very long proboscis. The possession of such a proboscis by our own bees would add millions of dollars annually to the wealth of the United States.”

CANADIAN DEPARTMENT.

R. F. HOLTERMAN, EDITOR.

We have to report a very heavy loss; at first one-third dead was supposed to cover the ground, but later reports will probably bring it nearer 50 per cent. The losses may briefly be accounted for, at least largely: by the failure of the honey crop after clover; consequent cessation of breeding and old bees ready for winter quarters; an unusual consumption of stores in the fall, followed by a severe winter. Men who stimulated breeding until a proper season, fed sufficient stores of the right kind and at the proper time, and properly packed their bees outside or had them in good cellars, were generally successful. Experienced beekeepers have generally been successful as they know and are careful about getting the proper conditions for successful wintering.

Martin Emyh, of Holbrook, wintered 178 out of 180; one winter he lost none out of 150 and his success is fully as great in summer management. He doubtless could give much information of value but we seldom hear of him in public.

Most parties report bees to have died in March after the severest weather had passed. Is this not additional proof that loss was caused by aged bees going into winter quarters and becoming worn out, and insufficiency of stores? One report before me gives eleven starved out of thirteen. Bees in some localities are doing remarkably well this spring. Fresh honey is being stored in abundance, combs having honey from top to bottom, and here we feel confident some of our colonies, unless there is a sudden change in the weather, will require extracting before the end of the week and the prospects are that beekeepers will reap an abundant harvest.

Brantford, Ont.

BOOK NOTICES AND REVIEWS.

—“Money in Potatoes” is the title of a valuable and instructive little manual published by the Franklin News Co. of Phila., Pa., which they kindly sent us for review.

—Mr. A. I. Root of Medina, Ohio, has also sent us a potato book entitled the A B C of Potato Culture, written by T. B. Terry of Hudson, Ohio, which is well illustrated and printed in good clear type.

Mr. Henry Alley, of Wenham, Mass., has submitted for review, the proof sheets of the latest edition of the “Beekeeper’s Handy Book.” Our reason for not waiting until the work was complete was because it will be placed on the market at about the same time that our subscribers receive their journal.

Mr. Alley’s former work was an invaluable addition to bee literature, giving as it did, not only the most scientific, systematic and practical method of raising queen bees, but also many other items of interest and much valuable information, gleaned from an intimate acquaintance with beekeeping of about twenty-seven years. The present work, however, surpasses the former in every respect, and no beekeeper can afford to do without it.

The instructions given have been thoroughly tested by the author who has made a success of the business, and these instructions are presented to the reader in a simple matter-of-fact way: They are, in fact, the concise description of the every-day experiences, for nearly twenty-seven years, of a thoroughly practical apiarist who has mastered every branch of the business.

This work contains about 300 pages, and nearly one hundred illustrations.

While we have other valuable works on Apiculture, yet none of them fill the position occupied by the “Handy Book,” and no matter what works one

may have this one should be procured, as it will more than return its cost each season.

The first part contains about 175 pages, is devoted to general subjects of apiculture, while the second is devoted to queen-rearing.

The author, so to speak, takes the beginner out into his apiary and gives him just the information needed to enable one who is almost wholly unacquainted with the bees, to commence this study and continue it with a surety of success.

He tells the beginner “how to start an apiary,” how and when to select, pack and remove their bees, when to place the section-boxes on, how to prevent bees from deserting surplus boxes on cool nights, and how to strengthen weak colonies.

His description of swarming and the proper care and management of swarms is excellent, while that pertaining to queenless colonies and their care is very important and valuable.

The subject of comb-foundation and its manufacture is complete and exhaustive, and is written by one of the best manufacturers of comb-foundation in the country.

The chapter on beehives, their construction, etc., is entirely a new departure, and original with this work. A large number of the best are illustrated and described, also their valuable features, as claimed by their originators.

The work also contains a brief, but interesting, description (with fine illustrations) of the various honey producing plants, and also the enemies of bees.

The second part, which is devoted to queen-rearing, has been revised and corrected, and many new and valuable illustrations and items of instruction have been added.

It is a work that we can heartily endorse and recommend as indispensable to every beekeeper.

NOTES AND QUERIES.

—Mr. J. M. Shuck who a short time since suffered a severe loss in the burning up of his beehive stock, has just sent us his neat and attractive circular stating that he is again prepared for business.

—We have just received from Mr. Pryal of Temescal, Cal., a complete file of the *California Apiculturist*, which we are pleased to add to our collection. Mr. Pryal will please accept our thanks for the same.

—We shall issue our Journal from June to December on the 15th of the month.

—Those who are so strictly orthodox in their opinions, especially when criticising others, should remember that their first duty is to practise what they preach, as their teachings will then be productive of good.

—We shall continue to issue our Journal from the Salem Press office, but all communications should be addressed to Wenham, Mass., and when sending money do not forget to make all postal notes or money orders payable on Salem, Mass., post office.

—We have just received a circular and price list from Mr. B——, who is now in Europe dealing in foreign queens and bees. The character of the language and criticisms contained therein will not warrant us in giving it a favorable notice. Manly criticism or counsel and advice, or one's opinion expressed in kindly language, always demands respect; but since Mr. B—— has engaged in the sale of the eastern bees and queens he has introduced some finely drawn lines of distinction between the names given to the different races of bees from the Holy Land.

Until lately we had no trouble in understanding what was meant by the terms Holy Land bees, Syrian bees or Palestine bees, and in order that we may be set at rights again will some one *capable of judging* kindly tell us

what countries are included within the boundaries of the Holy Land (not Holy Land proper.)

Mr. B—— is not even contented to establish names of his own but makes it a point to abuse all those who differ from him.

We trust that he may yet learn to allow others the same freedom in expressing their opinions that he wishes himself and then abstain from abusing them for so doing.

—We have just received from F. D. Wellcome of Poland, Maine, the finest lot of red raspberry plants that we ever saw, they certainly do him credit.

—As we are now publishing 5,000 copies per month of our Journal, advertisers will do well to give it a trial.

—Use every effort to work up a home demand for your honey this season as in most cases it will pay much better than sending it to city markets.

—Read all our club offers for this month carefully and if you should receive more than one copy of our Journal kindly hand the other to your neighbor. It may do him some good and it will help us.

Better chances were never offered to secure first-class goods cheap than will be found in our club list.

—Mr. J. C. Clark submits his communication to us for consideration, and we would state that while the extractor in the hands of an inexperienced apiarist or by injudicious use will doubtless prove a great injury, yet we consider it one of the most valuable implements in a well conducted apiary.

The trouble with the “glutted” honey market comes, not from the use of the honey extractor, or the over-production of honey, but from a lack of proper knowledge and means of creating a demand. As a rule we do not consider it best to extract honey from combs containing unsealed larvæ, but one must be governed by his knowledge of and experience in this business.

We could not be hired to do without the extractor for many reasons.

We should be pleased to hear from Mr. Dadant on this subject, as he uses the extractor extensively.

A few days since while we were busy at the desk, Mr. Alley called us out into the apiary to witness a swarm of bees issuing from a hive that was provided with one of his drone traps. It was very amusing and interesting to watch the honey-laden bees rush pell-mell out through the perforated metal entrance giving conclusive evidence that the perforations were amply large to freely admit a worker bee laden either with pollen or honey.

Soon the queen unable to force her way through at the entrance passed into the trap. As soon as the majority of the bees had passed out we removed the drone trap, fastened it to the end of a long pole and rested it against a small tree near by. In a short time the bees, discovering their queen in the trap, clustered upon it.

When the bees had become quiet we removed the trap to a convenient shady spot near by and laying it on its side placed over it one of the swarming boxes, covering the box with a board so as to cover the wire cloth and make it dark within the box.

We then left the swarm in that condition for about six hours in order to experiment with it and learn whether the queen would force her way out.

In the early evening we examined the swarm and found that the queen was out among the bees in the swarming box, and found that as the trap was laid on its side the bees that filled the trap formed a bridge over which the queen could pass through the wire tube through which she passed into the trap.

This is a valuable feature of the trap because if one cannot always be at hand to attend to the swarms, the queen after about six hours will find her way back into the hive again, thus preventing either her loss or that of the bees.

We have had considerable experience with hiving swarms, but never performed the task with such ease or so speedily as we can with the drone trap. We can heartily endorse it as invaluable to the beekeeper even for this one purpose alone.

We shall experiment further with the trap and give the results.

—Mr. J. D. Goodrich of East Hardwick, Vermont, has sent us some most excellent comb foundation. The thin for sections was especially fine.

—For \$3.00 cash we will send the “American Apiculturist” one year and one of our fertilizing hives containing four combs with brood and one-half pound of bees and a choice selected queen—the hive will also be supplied with one of our cone feeders. With this small nucleus colony one can rear a number of queens during the season and it will be a great help in learning queen-rearing.

—Advertisers will please notice the change in our rates, 15 cents per line each insertion. There are about seven words to the line, and twelve lines to the inch. In sending copy always write it on one side of a separate sheet of paper and be very particular to have the names, addresses, etc., very plain.

Parties unknown to us must send cash with the order.

All copy for advertisements and manuscript must reach us by the first of the month, if they are to appear in the following number which we shall issue on the 15th of the month until December.

LETTER BOX.

Des Moines, Ia.

DEAR SIR:

Am in receipt of Vols. 1 & 2 of the "American Apiculturist" handsomely bound in cloth with an enormous golden bee on the front cover. Happy bee! to thus repose upon a production that has sprung up under the skies of Quinby, Carey, Parsons, Elwood, Alley, Hetherington, Doolittle, L. C. Root, and a host of northeastern beekeepers who have honored the calling.

The attractiveness of the book and its practical teaching should find for it a ready sale.

J. M. SHUCK.

Willamina, Yarnhill Co., Ore.
April 22, 1885.

S. M. LOCKE, SIR:

I have just received your first and second volumes bound together in cloth, and must say that it exceeds my greatest expectations. Every beekeeper should have it.

FRANK S. HARDING,

Pres. of the Willamette Valley B. A.

Oxford, Butler Co., Ohio.

DEAR SIR:

Vols. 1 & 2 of the "Apiculturist" received, and after carefully examining its contents, I have no hesitancy in pronouncing it fully up to any bee journal published. Its contributors are first-class, and show by their articles that they are men of ability. I highly approve of your treatment and condemnation of fraud in every shape. I am pleased to meet with the familiar face of our old friend Mr. Langstroth in Vol. 2. He lives within 200 yards of me, and is one of the dearest old men I ever knew. A person has only to know him to appreciate his worth.

D. A. McCORD.

Battle Ground, Ind., April 13, 1885.

Dear Sir:

The "Am. Apiculturist" came safe to hand, and having examined its contents, or rather the headings of the major part of the book, I pronounce it *ne plus ultra*. It should be in the hands of every respectable beekeeper of our broad land, it being complete and full of most useful knowledge. With many thanks I am, sir, very truly yours in fraternal bonds of apiculture.

P. S. I truly wish you success, in all that is good and honorable, and shall try to furnish an occasional article for the Am. Apiculturist. Please find this my second contribution to same. If acceptable you are at liberty to publish. Trusting the "Apiculturist" may live long and prosper, I am as ever,

J. M. HICKS.

Brantford, June 4, 1885.

FRIEND LOCKE:

I see by the last number of the "Apiculturist" much new and valuable information from England and Germany. Last fall I learned through a German journal that the poison of the honey bee was utilized in curing honey.

A thought suggested itself to me which I hoped ere now I should have been in a position to practically test, but circumstances have been unfavorable

and perhaps you know of some one who would take an interest in the matter.

I intended taking a strong colony and feeding them, say 3 lbs. of syrup per day and allow the bees to seal this and extract with great care. Then feed the same colony 25 lbs. in a day of twenty-four hours (and it can be done) and after it has been sealed extract as before, during hot weather, and again later in the season, giving four distinct experiments, and then test the various syrups for difference, if any, of quantity of poison contained in stores. If a marked difference occurred after repeated experimenting the matter might be worthy of consideration and thus one reason why stores put in late are not as good as those given earlier.

I must say a man must be too much of an enthusiast if he considers, as many appear to, that any one condition, if fulfilled, will insure perfect success in wintering. Doubtless much is to be learned in wintering, and theories advanced will often be another step towards success. If we only acted thoroughly up to our present knowledge and were more careful, the percentage of loss would doubtless be wonderfully reduced.

Your brother beekeeper,
R. F. HOLTERMAN.

We are pleased with Mr Holterman's suggestions, and, if possible, will make some tests, as this matter is of more vital importance to beekeepers than they are aware. We shall refer to this again in the near future.

Mr. Holterman has touched the right key; we must have more careful experimenting with a view to solving some of the mysteries that now surround apiculture, and we trust that others of our readers and beekeeping friends will offer like suggestions and also assist us in making the tests. It will pay you well to do so, and we can by this means help each other very much.—ED.]

FOOTNOTES:

- 1 Read at the Beekeepers' Congress, at New Orleans.
- 2 If our friend will visit us we will show him colonies of Italians that can be examined without fear of stings, and this with neither smoke nor protection.—ED.
- 3 When working for comb honey, I return all of my swarms so that I get no increase that season.—D. F. L.
- 4 Doubtless our readers would be pleased to hear from friend Lashier again, and we trust that he will favor us with the article.—ED.
- 5 Nature has wisely ordered it that generally only one-third of the bees in a colony fly out to the fields, so that the colony would not perish even if all the absent bees were lost through some mishap.
- 6 Literally magazine or wave house hive.—R. F. H.
- 7 We are unable to decide to what extent the "Chaff Hive" was liable for the loss of our friend's bees, as quite frequently the lack of a thorough knowledge of beekeeping will result in loss of queens or bees which is attributed to the hive or dealer.

Our friend certainly has a right to speak and the columns of the "Apiculturist" are always open to those who wish to express opinions, so long as they do so in a fair, manly way.—ED.
- 8 By mistake this communication, which should have appeared in our last, was overlooked and as it contains valuable items we give it in this number.—ED.
- 9 This paper was read at a meeting of the New York Microscopical Society, March 6, 1885.
- 10 The ocelli are simple eyes or lenses, set between the compound eyes, as additional organs of vision. These organs are possessed

by all insects which have compound eyes.—ED.

Transcriber's Note:

Obvious printer errors corrected silently.

Inconsistent spelling and hyphenation are as in the original.

*** END OF THE PROJECT GUTENBERG EBOOK THE AMERICAN
APICULTURIST. VOL. III. NO. 6, JUNE 15, 1885 ***

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